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Terminalia bellirica Fruit Extract Mediated Synthesis of Gold Nanoparticles (AuNPs) and studies on Antimicrobial and Antioxidant activity

Viswadevarayalu Annavaram^a, Venkata Ramana Posa^a, Vijaya Lakshmi D^b, Sumalatha J^c, Adinarayana Reddy Somala^{a,*}

^aDept of Material Science & Nanotechnology, Y.V. University, Kadapa, Andhra Pradesh- 516 003, India

^bDept of Microbiology, Yogi Vemana University, Kadapa. Andhra Pradesh- 516 003, India

^cDept of Pharmaceutical Chemistry, Nirmala College of Pharmacy, Kadapa, Andhra Pradesh- 516002, India

*anreddyphd@gmail.com

Abstract

In the current report, we presented a biological and eco-friendly method for the synthesis of gold nanoparticles (AuNPs) by *Terminalia bellirica* (TB) fruit extract. Fourier transform infrared spectroscopy (FTIR) indicates that –OH functional groups were involved formation of AuNPs. X-ray diffraction analysis revealed that the biosynthesized AuNPs exhibit high crystalline form and spherical shape with an average size is 20-30 nm. The zeta potential of the compound was found to be -58.0 mV. AuNPs were found to be effective against *Candida tropicalis*, *Candida albicans* isolated from clinical samples, AuNPs also effectively worked as free radical scavenging activity.

Keywords

Green synthesis, Gold nanoparticles (AuNPs), Antifungal activity, Antioxidant activity.

Introduction

Nanotechnology emerges as a potential technology on the future by knocking all the available doors of science and technology. The synthesis of nanoparticles is one of foremost vital part of the nanotechnology and its application. Synthesis of metallic nanoparticles has fascinated considerable attention due to their remarkable physical and chemical properties such as optical, electronic and catalytic properties [1-3]. As a result, nanoparticles have useful applications in optical and biological sensors, conductive materials, and photovoltaic cells [4]. The production of noble metal nanoparticles such as Au, Ag, Pd, Pt etc., is of great interest due to their unique properties. Among these noble metal nanoparticles AuNPs are subjected to scalable and dispersion processes and these nanomaterials offer great scope for new research in the field of nano crystal synthesis. AuNPs are having wide variety of useful properties for applications in pharmacology, medicine and food industry etc., and they possess various biological applications including labeling, drug-delivery, photothermal therapy, tissue/tumor imaging [5]. For the synthesis of AuNPs so many methods have been improved such as chemical reduction [6], solvo thermal [7], sonochemistry [8], micro emulsion techniques [9], laser ablation [10], polyol method [11], DC arc discharge method [12] and green synthesis [13]. Between these methods green synthesis is simple, less expensive, low-energy and low-hazard aqueous method.

In recent years, use of biological samples such as plant parts, algae, microbes and bio molecules for the synthesis of biosynthesized nanoparticles has been emerging as promising environmental friendly methods. In ancient days also particles were synthesized in ecofriendly method. For that one of the example is that, gold was used in the form of gold ash in Indian ayurvedic/native medicine, and we called it as globular gold particles size upto 56-57 nm [14].

Recently, gold nanoparticles synthesized using plant samples, *Nerium oleander* [15], *Pogestemon benghalensis* (B) O. Ktz [16], *Moringa oleifera* [17], *Cassia auriculata* L. [18], *Salvadora persica* [19], *Murraya koenigii* Spreng. [20], have been reported.

Terminalia bellirica is Combretaceae family, large deciduous tree found throughout India. It is commonly known as "Bahera" or Beleric or bastard myrobalan. The main phytochemicals present in fruits are alkaloids, phenol, tannins and flavonoids [21]. Plant parts of *Terminalia bellirica* have enormous medicinal properties. The native medicates recommends this fruit in powder form or liquid form for many infectious diseases. In *Terminalia bellirica* fruit extract contain beta-sitosterol, gallic and ellagic acids, ethyl gallate, galloyl glucose, chebulagic acid and a cardiac glycoside. But here 21.4% pseudo tannins were present [21] so here gallic acid is a pseudo tannin. Below these phytoconstituents chemical structures were included.

This work aims to add AuNPs to the *Terminalia bellirica* fruit extract mediated nanoparticle synthesis. The nanoparticles were characterized with spectroscopic techniques and HRTEM methods, including in situ elemental analysis. The zeta potential measurement gives stability of the synthesized nanoparticles. The antifungal activity and antioxidant activity of AuNPs is also observed.

Experimental

Materials

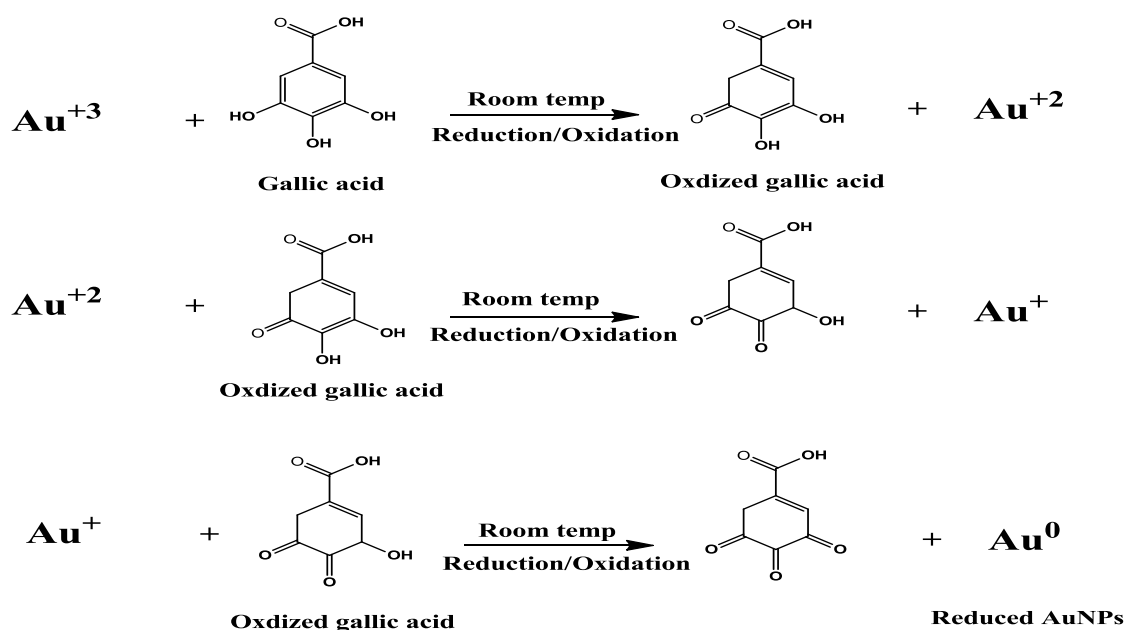
The Fresh fruits of *Terminalia bellirica* were collected during rainy season in the month of August from Palakondalu hills, Kadapa, Andhra Pradesh, India. Tetra-chloroauric (III) acid

(HAuCl₄.3H₂O) was purchased from Molychem Corporation, India; throughout the experiment double distilled water was used.

Methodology

Green Synthesis of AuNPs

TB fruits were surface cleaned, dried at shade, crushed and grinded with mortar and pestle, 5 g of pulverized fruit powder was boiled for 20 min, this crude extract is filtered and stored at -4 °C for further experiments. The aqueous solution of 0.1mM HAuCl₄.3H₂O was standardized in 100 ml of standard flask. 10 ml of the plant extract was mixed with 90 ml of tetrachloroauric (III) acid solution under magnetic stirring at room temperature for the reduction of Au(III) ions into Au(0), until the color of the mixture changed from yellowish to wine red coloration. The appearance of wine red coloration shows formation of AuNPs. The synthesized AuNPs were purified, collected and stored properly for future uses. We represented a probable mechanism for the formation of AuNPs.



Probable reaction mechanism involved for the green synthesis of AuNPs.

Characterization of AuNP

UV–Visible spectroscopy analysis

After reduction, the reduced AuNPs in aqueous solution was found using Shimadzu UV-1800 model of UV-spectrometer at different time intervals. The resultant samples UV-spectrum were recorded using quartz cuvettes at a 1 nm resolution as a function of reaction time.

Fourier transform infrared spectroscopy (FTIR) analysis

The reduced gold nanoparticle solution is purified with centrifugation to complete removal of secondary metabolites present in the solution. The centrifuged compound is dehydrated at 60 °C. The 20 ml of fruit extract is dried with rotary evaporator by using Rotavapor R-215, and then after the compound is washed with methanol to dissolve dried compound in round bottom flask and the collected suspension is dried at 50 °C. The dried compound of AuNPs and *TB* fruit extract is characterized with FTIR spectrum. The sample was recorded by Perkin-Elmer, Two Model FTIR spectrometer. The FTIR ranged from 4000 to 500 cm^{-1} at resolution of 4 cm^{-1} by making a KBr pellet with 2 mg AuNPs and dried *Terminalia bellirica* fruit extract. The resulted data values compared with fruit extract and AuNPs, which functional groups involved in the formation and capping of gold nanoparticles.

Powder X-ray diffraction

Purified AuNPs were characterized by powder X-ray diffractometry (XRD) Bruker D8 using $\text{CuK}\alpha 1$ (1.5406 Å) and $\text{K}\alpha 2$ (1.54439 Å) radiations. The XRD pattern was analyzed to establish peak intensity, position, and width.

High resolution transmission electron (HRTEM) analysis

The size and shape of the biosynthesized nanoparticles were observed by HRTEM measurements using JEOL 3010 instrument operating at 200kV. For HRTEM analysis were prepared by placing a probe sonicated drop of nanoparticle solution on a copper grid and allowing dispersant to evaporate. During HRTEM imaging EDS is also performed for elemental analysis by using (SUPRA 55) - CARL ZEISS instrument operating at 5 to 20 kV.

Dynamic light scattering (DLS) and zeta potential (ζ) analysis

By using dynamic light scattering method characterization of particles dispersed or dissolved in a liquid. Particle size and zeta potential of AuNPs was measurement using HORIBA SZ-100. The dispersed solution is taken into quartz cuvette which is given with the instrument and analyzed at laser light to be scattered at different intensities. Zeta potential is a measure of the magnitude of the electrostatic or charge repulsion/attraction between particles, and is one of the fundamental parameters known to affect stability.

Antifungal activity

Disk diffusion method is used to determine inhibition of fungal cultures at different concentrations. Fungal test cultures of *Candida tropicalis* and *Candida albicans* was commercially purchased microbial type culture collection. For the growth of fungal cultures Sabouraud dextrose agar (SDA) is used in 10 g per liter double distilled water. Into this fungal suspensions different concentrations of (10, 20, 30, 40 μ g) of AuNPs added on middle of discs with micro pipette to diffuse. In other culture plate is used as control, here DMSO used as control and these culture plates incubated at 37 °C to observe maximum zone of inhibition.

Antioxidant activity

ABTS and DPPH free radical scavenging activity

AuNPs were tested for free radical scavenging activity for 2,2-diphenyl-1-picrylhydrazyl (DPPH) and 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulphonic acid) (ABTS). In many applications, to estimate the antioxidant capacity of food or their extracts, the stable nitrogen-centered radical DPPH has been used [22]. 0.5 ml of 1.5 mM DPPH was prepared and it is mixed with different concentration of AuNPs (0.25, 0.50, 0.75, 1.0, 1.25 mM/ml) incubated in dark for 30 min. After incubation the absorbance of the sample was characterized by UV-visible spectroscopy at 517 nm. Citric acid used as reference control for DPPH antioxidant activity.

For the testing of ABTS scavenging activity, 7.0 mM ABTS salt and 2.45 mM K₂SO₄ were mixed and kept in dark condition for 16 h, thereafter; it was used for the production of ABTS⁺. After incubation AuNPs mixed with this solution at different concentrations (0.25, 0.50, 0.75, 1.0, 1.25 mM/ml), the absorbance was characterized with UV-visible spectroscopy at 734 nm. ABTS ethanol reagent without sample was used as control.

Results and discussion

(Fig. 1a) gives a well-defined absorption peak at 525-530 nm [23-25] exhibited that the formation of colloidal solution of AuNPs and linear absorption spectrum of *TB* fruit extract. The colloidal solution of gold ion color changes from yellow to wine red color by the addition of *TB* fruit extract indicated that the conversion of Au(III) to Au(0). The (Fig. 1b) shows that the AuNPs were observed that the peak at 550 nm at different timeintervals (like 9, 30, 60, 90, 120, 150 min). In (Fig. 1b) indicates that when time increases absorption intensity is also increases because when reaction time is dependant for the reduction of Au ions into nanoparticle.

Fourier transform infrared (FTIR) spectroscopy is a primary tool to understanding which functional groups that are involved in the nanoparticle reduction and apparent oxidative stability

and to compare the *Terminalia bellirica* fruit extract to AuNPs (Fig. 2a). The absorption bands at 3391, 2926, 1615, 1447, 1209, 1350, 1039 cm^{-1} representing R–OH, R–CHO, $>\text{C}=\text{O}$, $-\text{CH}$, C–O, R–F, C–O respectively. Of all these bands indicates polyphenols, terpenoids, flavonoids and protein compounds present in *TB* fruit extract. The synthesized AuNPs showed significant peaks at 3436, 2924, 1633, 1052 cm^{-1} . This represents R–OH, R–CHO, and $>\text{C}=\text{C}$, C–O stretching vibration indicates different polyol compounds adsorbed on the surface of AuNPs.

The crystalline nature of AuNPs was further affirmed from X-ray diffraction (XRD) study. X-ray diffraction is used to portray crystallographic structure, particle size, and favored introduction in polycrystalline or powder solid specimens. This is a favored system for examination for portrayal of obscure crystalline materials. The XRD pattern (Fig. 2b) of powdered AuNPs showed Bragg reflections peaks at 38.14° , 44.24° , 64.64° and 77.39° in 2θ range in between 30° and 80° which can be indexed as (111), (200), (220) and (311) planes of AuNPs which are these planes relatively matched with reference value of (JCPDS file no.04-0784) [26] revealing that the synthesized AuNPs are composed of pure crystalline gold (Fig. 2b).

Figure.3a-c shows the TEM image of AuNPs at different magnifications. The AuNPs exhibit a spherical, triangular, hexagonal morphology but the maximum spherical shapes are visible with an average diameter of 20-30 nm. Further structural information was obtained from the high-resolution HRTEM image shown in (Fig. 3d-f). The size of synthesized AuNPs is determined by DLS method. It is shown inset of TEM image (Fig. 3a). Histogram revealed that particles obtained are poly disperse mixture with the size is ranges 6 to 60 nm. The mean particles size was found to be 20-30 nm. The lattice spacing in the image is consistent planes of the fcc phase of metallic Au, with a ~ 0.234 nm periodicity (Fig. 3f).

In (Fig. 4a) X-ray energy dispersive spectroscopy (EDS) conducted during HRTEM analysis for elemental analysis. The result represents the presence of Au in the samples and indicates that the phase-transferred nanoparticles have relatively less material associated with *Terminalia bellirica* than the as-synthesized sol.

The zeta potential is a key indicator of the stability of colloidal dispersions. The magnitude of the zeta potential indicates the degree of electrostatic repulsion between adjacent, similarly charged particles in dispersion. Figure.4b shows the zeta potential (ζ) of AuNPs, which is found to be -58.0 mV. AuNPs dispersed in double distilled water to measure zeta potential. Colloids with high zeta potential (negative or positive) are electrically stabilized. Here in this case high negative value confirms the repulsion among the particles and thereby increases in stability of the formulation [27].

Antifungal activity

The disc diffusion method is used to find out inhibition efficiency of green synthesized AuNPs. The 72 h grown fresh fungal culture was inoculated into the Sabouraud Dextrose Agar plate. 6 mm diameter Whatman No. 1 sterilized discs were placed on SDA plate for disc diffusion method. AuNPs were dispersed in DMSO solution. 5 μ l capacity discs with the concentration of the compound is designed which ranges from 10-40 μ g. The *candida* species were spread on to the medium and the discs were placed and various volumes of AuNPs (10, 20, 30, 40 μ g) were added at the center of the disc. 5 μ l DMSO is added for control at the center of the plate for the antifungal activity. The culture plates were incubated at 37 °C for 48 h in which the antifungal activity was evidenced by the presence of a zone of inhibition (mm) around the disc. The resultant zone of inhibition on culture plates is observed and the values showed in (Fig. 5a).

Antioxidant activity

Antioxidant activity of AuNPs was determined for DPPH scavenging activity. Here citric acid used as standard. DPPH radical scavenging activity was observed by color changes from violet to red color. The color of the solution changes when concentration increases. The scavenging activity increases with increasing concentration of the AuNPs. The ABTS radical generated from the oxidation of ABTS is an excellent tool for determining the antioxidant activity of the chain breaking antioxidants [28]. The ABTS quenching effect of free radical scavenging activity of the AuNPs increased from 20% to 40% with increasing concentration of AuNPs. By comparing with ABTS, DPPH scavenging activity shown higher scavenging activity. (Fig. 5b and c).

Conclusion

In conclusion, *Terminalia bellirica* fruit extract has been shown to reduce Au(III) ions into form AuNPs. By using FTIR analysis data of the *Terminalia bellirica* with AuNPs indicating that functional groups of fruit extract is responsible as reducing agent mixtures for the formation of nanoparticles and the hypothetical reaction mechanism represents that the formation of AuNPs. Analysis of HRTEM images shows that the synthesized AuNPs were mostly single crystalline and spherical shape. The histogram of particle size analyzer shows that the average diameter of AuNPs is 20-30 nm. The antifungal activity against *Candida tropicalis*, *Candida albicans* was also studied. We concluded that the functional groups of *Terminalia bellirica* fruit extract that reduces the gold nanoparticles and appear to protect them from further oxidation is ongoing, zeta potential value is also confirms that the particles have high stability. The DPPH scavenging activity is higher than ABTS scavenging activity.

Conflict of interest

The authors declare no competing financial interest.

Acknowledgements

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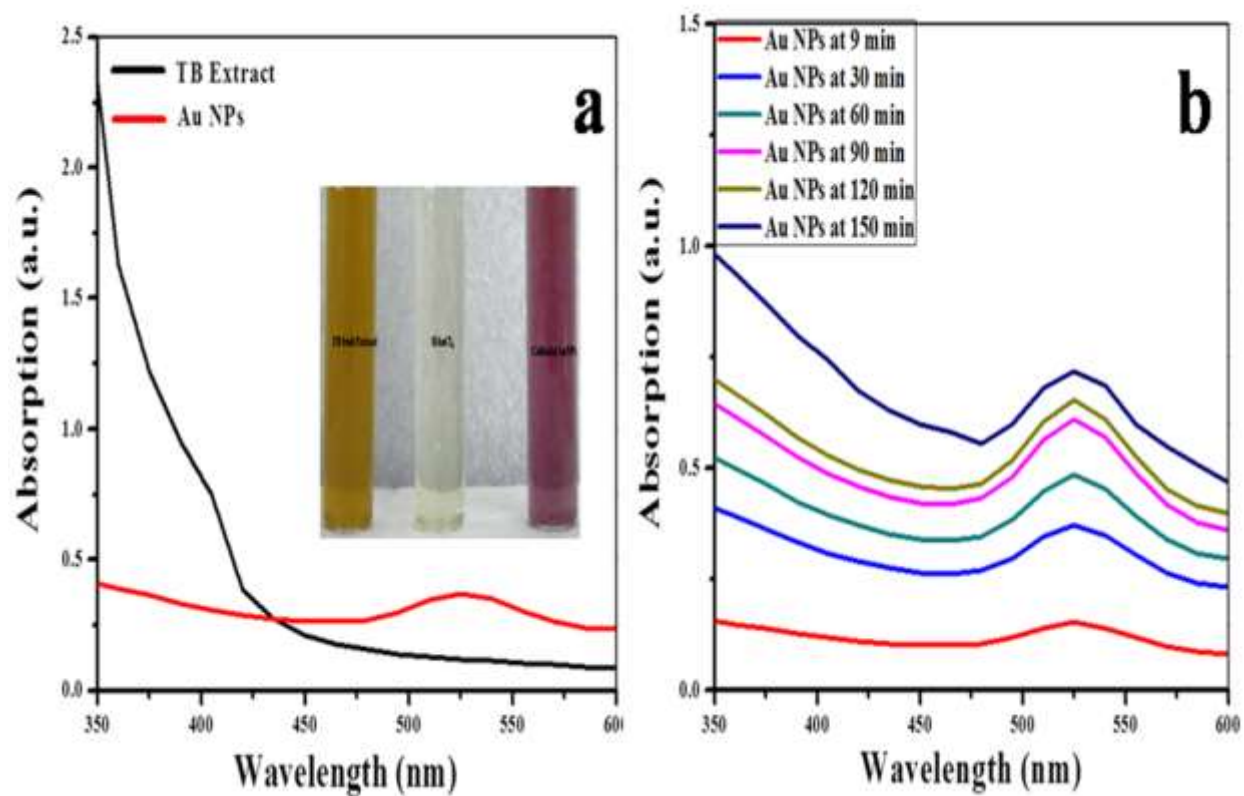


FIG. 1 (a) UV-Vis absorption spectra of extract and synthesized AuNPs. (b) UV-Vis spectra of the AuNPs prepared with $\text{H[AuCl}_4\text{].3H}_2\text{O}$ solution with *TB* fruit extract at different time intervals.

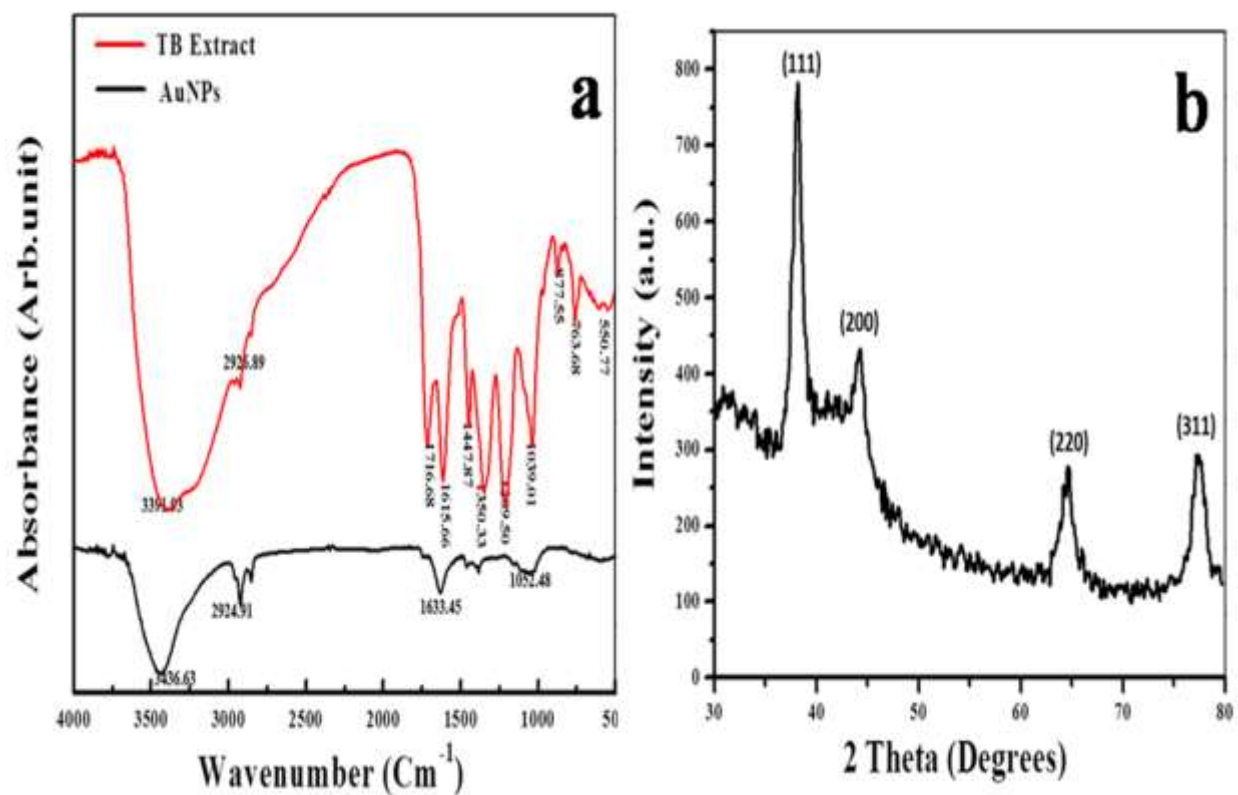


FIG. 2 (a) FTIR spectra of the *TB* fruit extract and the synthesized AuNPs. (b) XRD pattern of AuNPs.

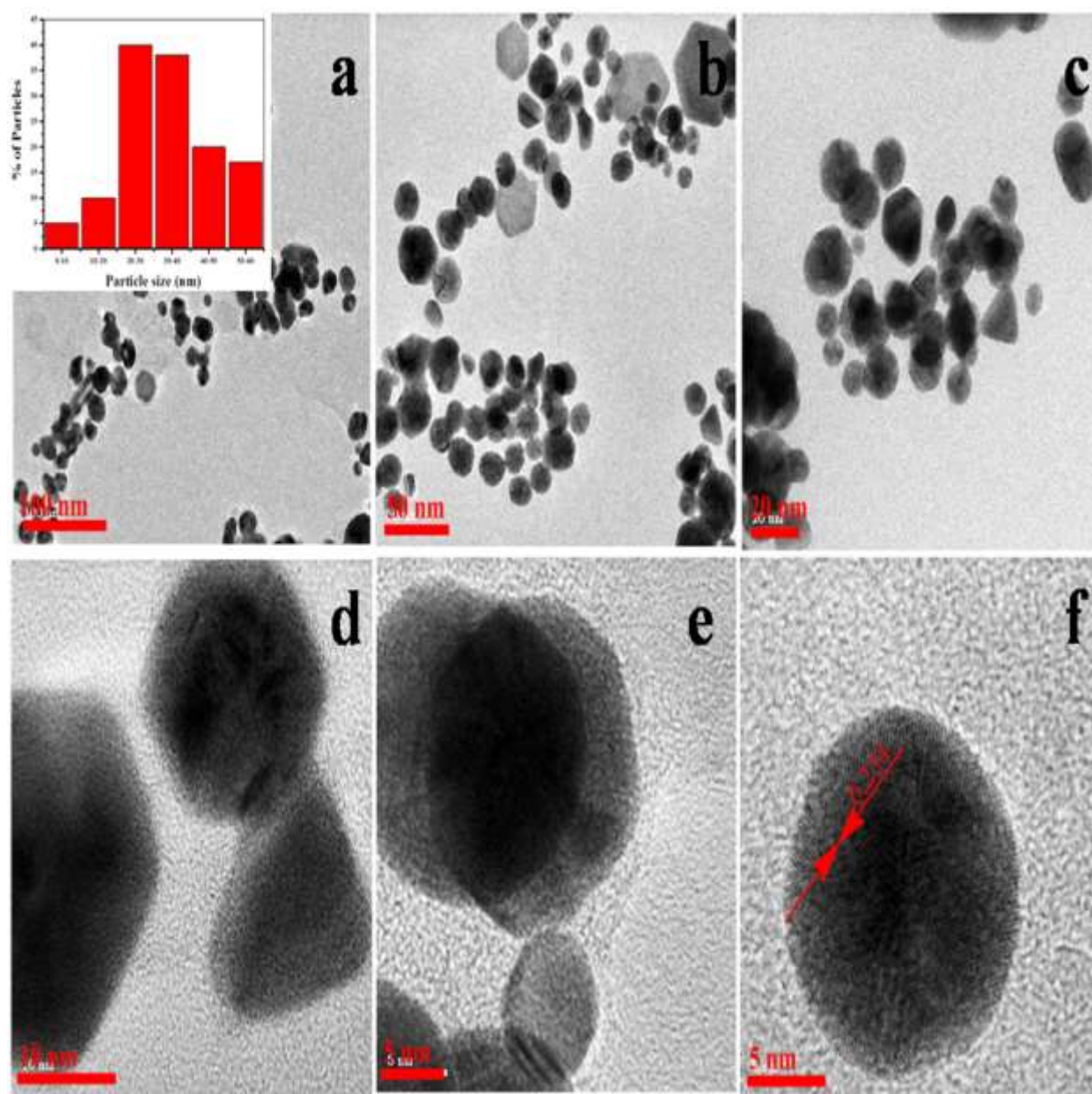


FIG. 3 (a–c) TEM images of the AuNPs at different magnifications prepared by *TB* fruit extract. Inset show the corresponding particle size distribution histogram of AuNPs respectively. (d–e) HRTEM image of AuNPs.

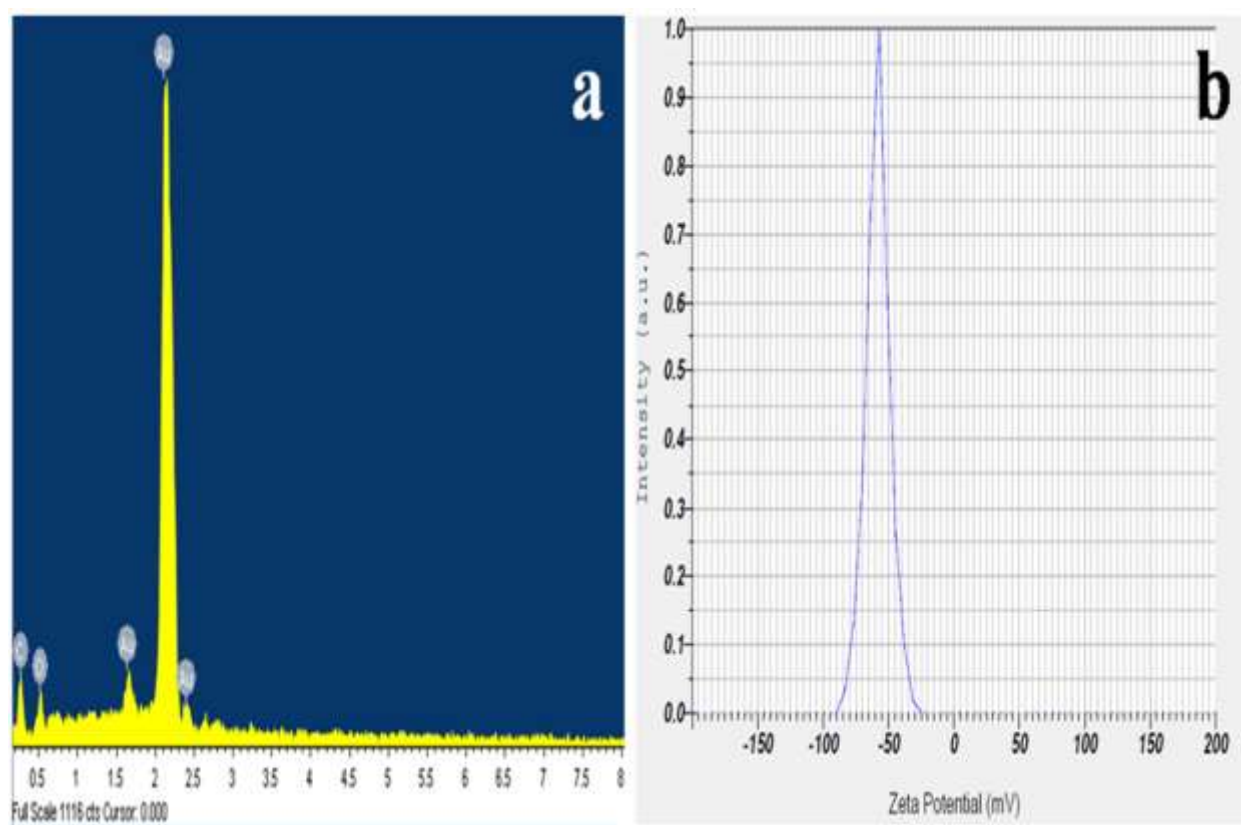


FIG. 4 (a) EDX spectrum of synthesized AuNPs. (b) Stability of AuNPs at -58.0 mV in zeta potential analysis.

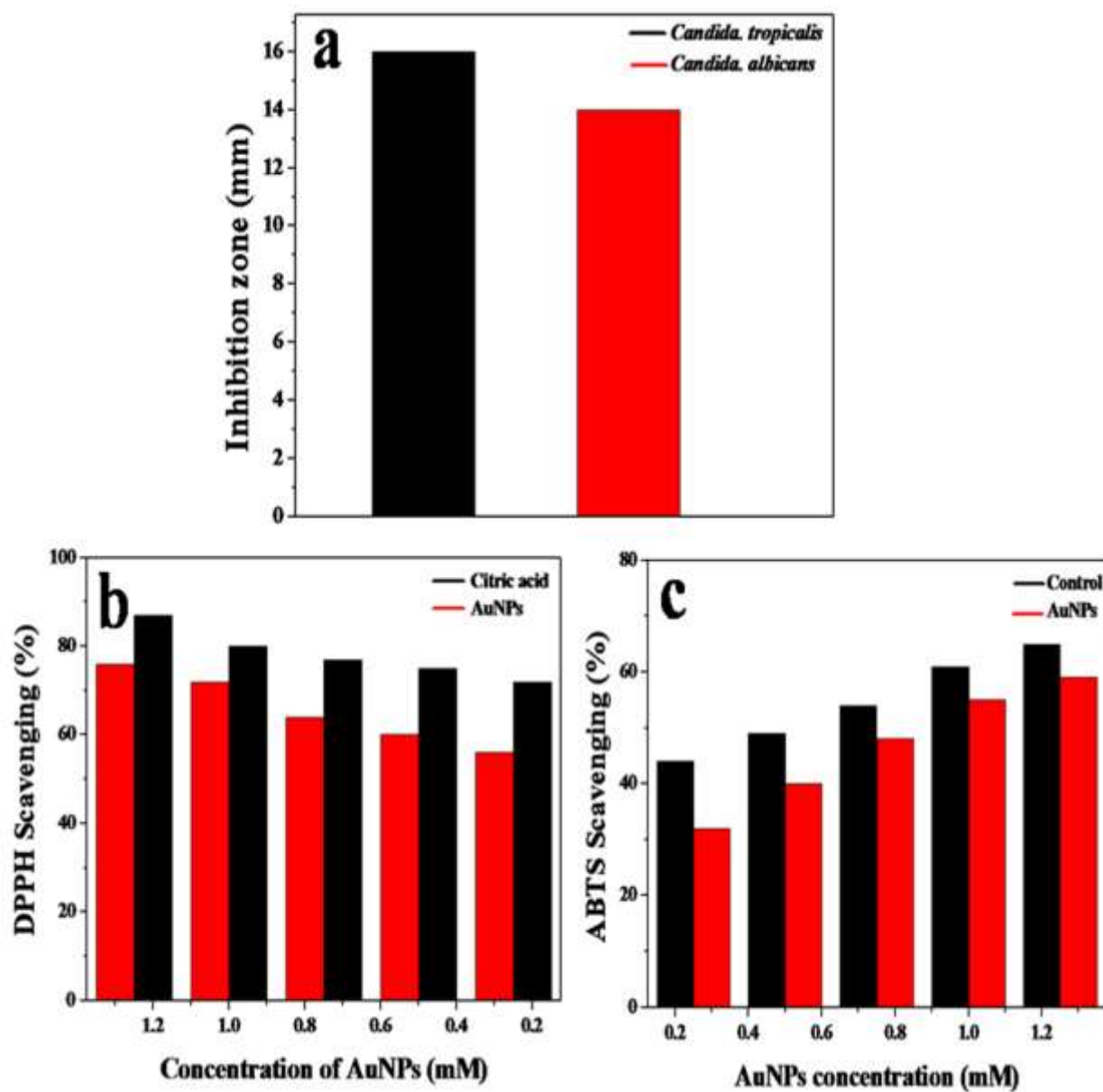
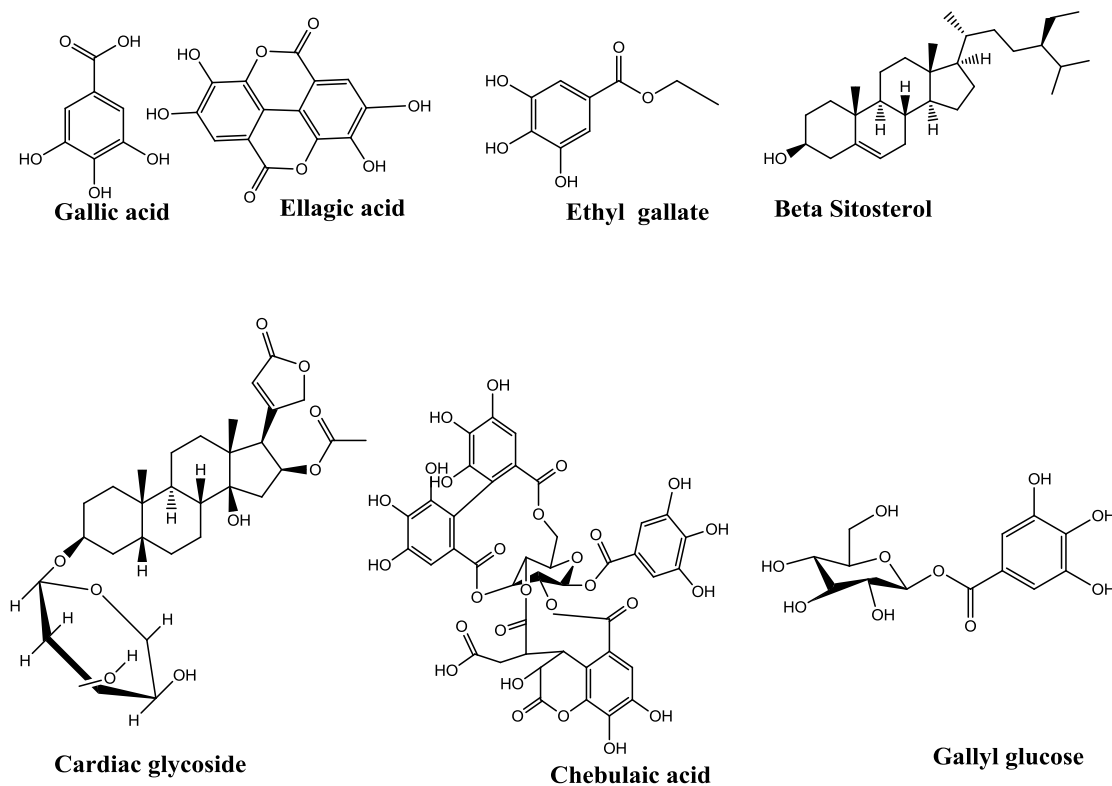


FIG. 5 (a) *Candida tropicalis* and *candida albicans* maximum zone of inhibition of AuNPs. (b) DPPH free radical scavenging activity of AuNPs at different concentrations with control. (c) ABTS free radical scavenging activity of AuNPs at different concentrations with control.



Scheme 1. Phytoconstituents present in the *Tb* fruit extract and their chemical structure.